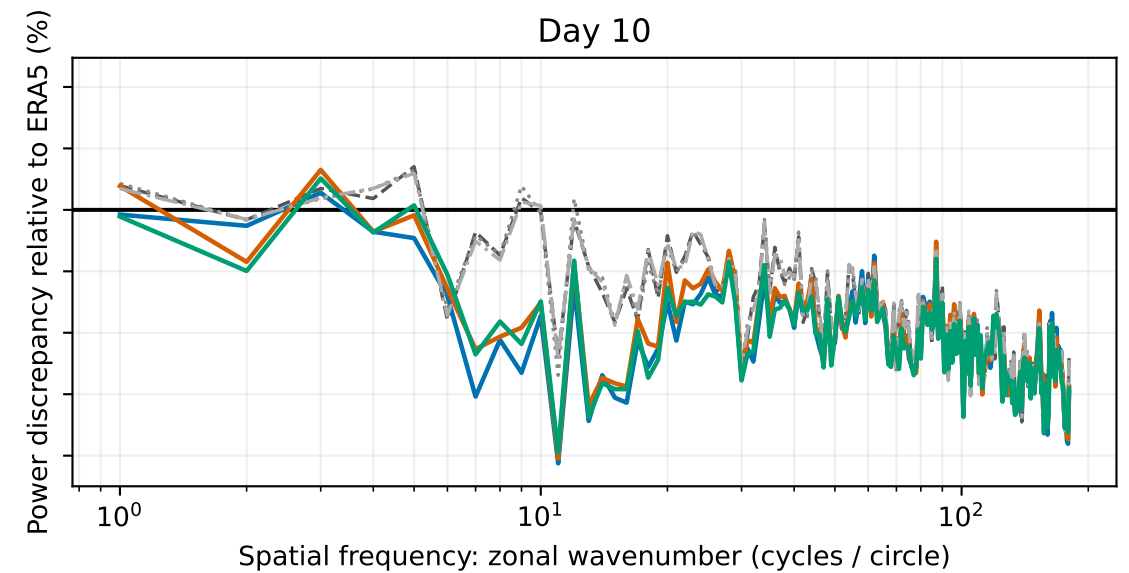
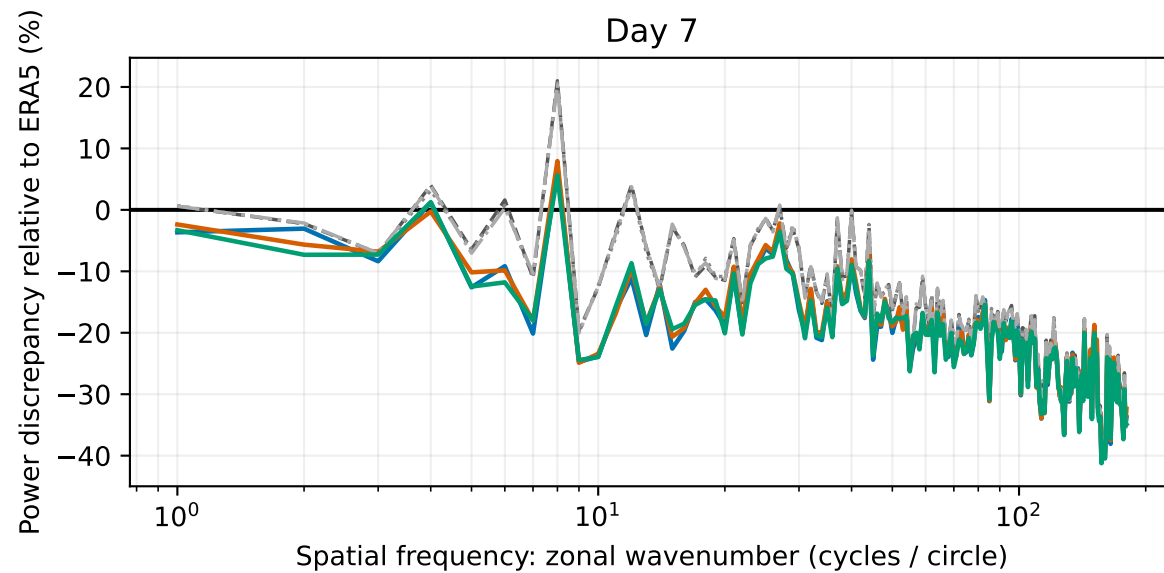
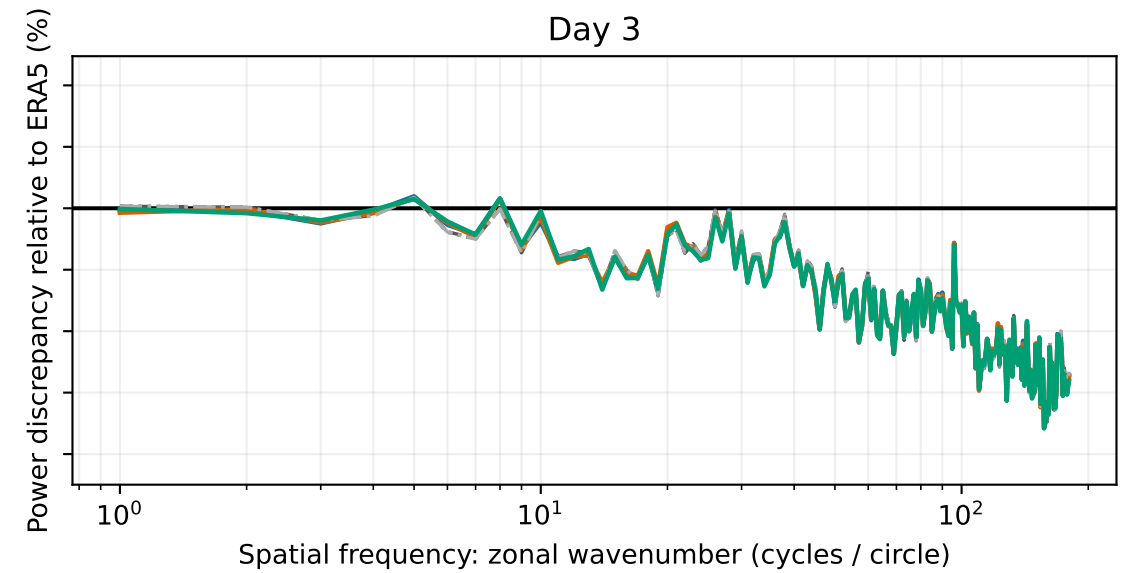
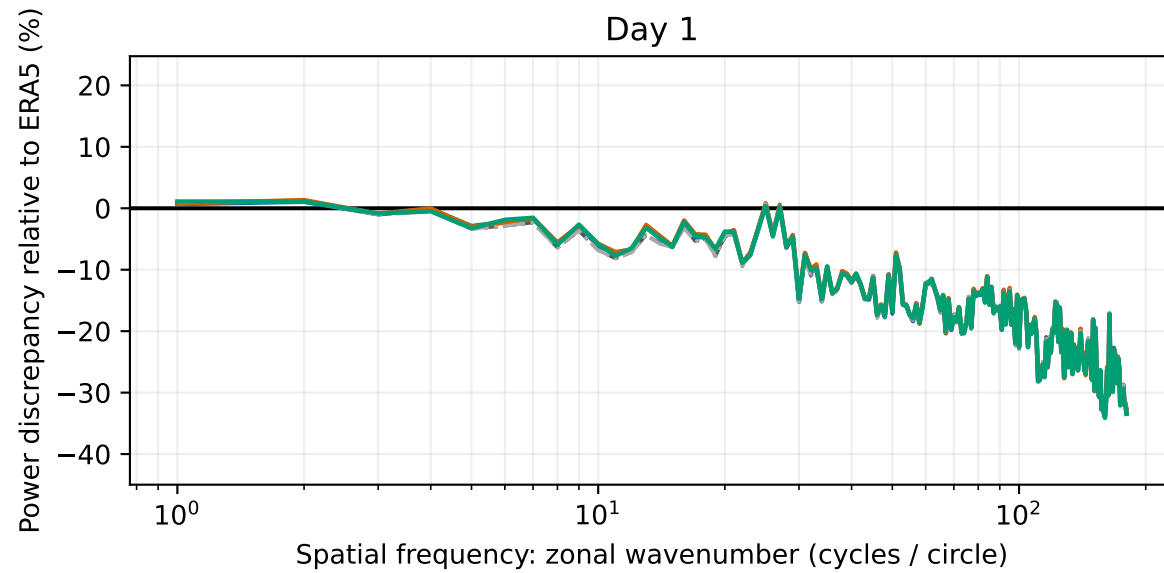


T2m: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power

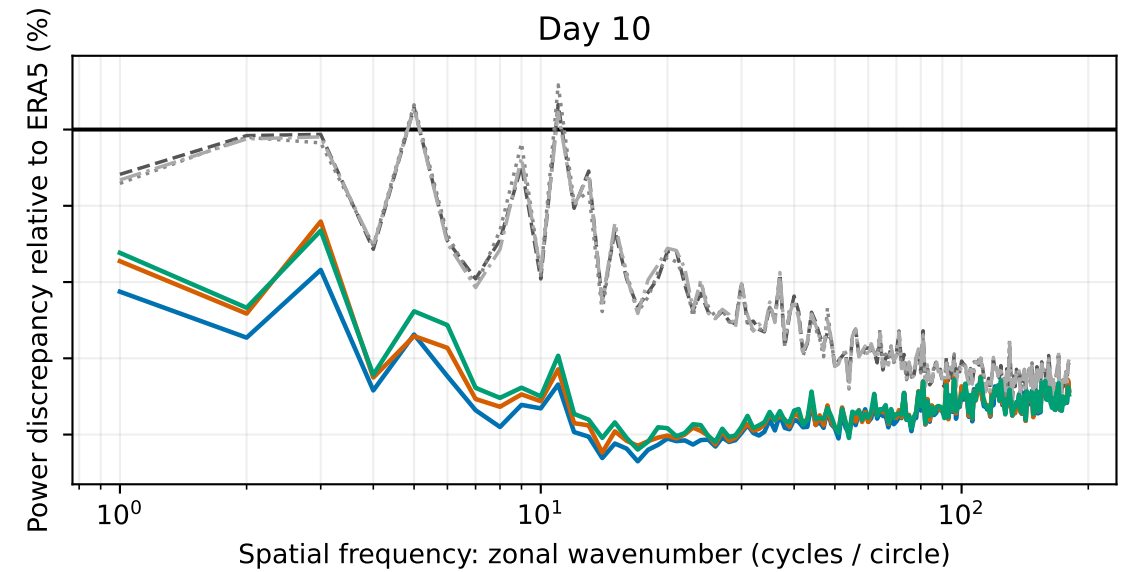
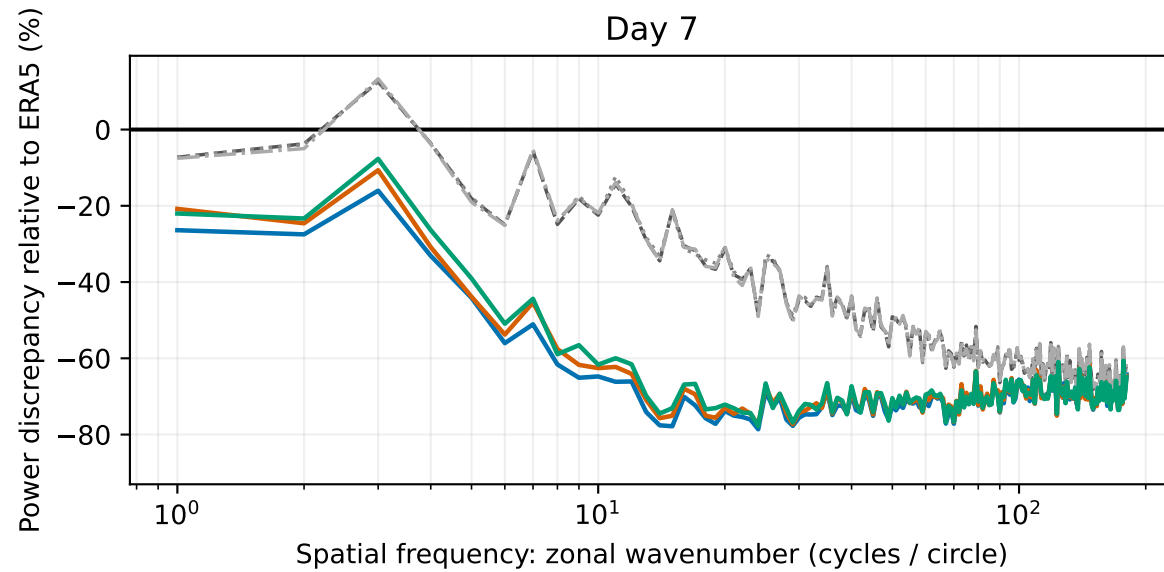
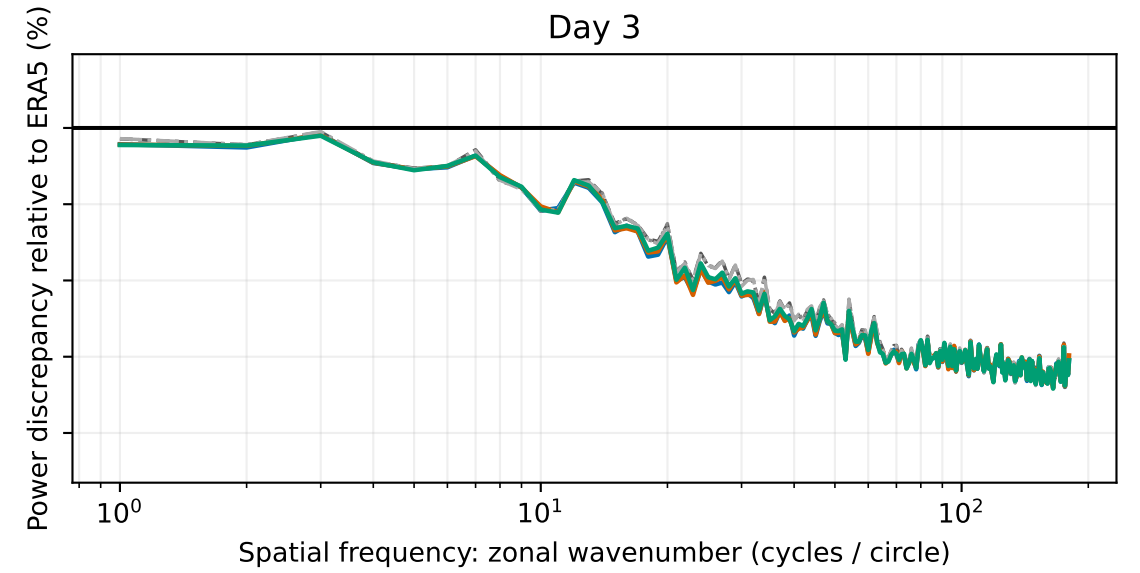
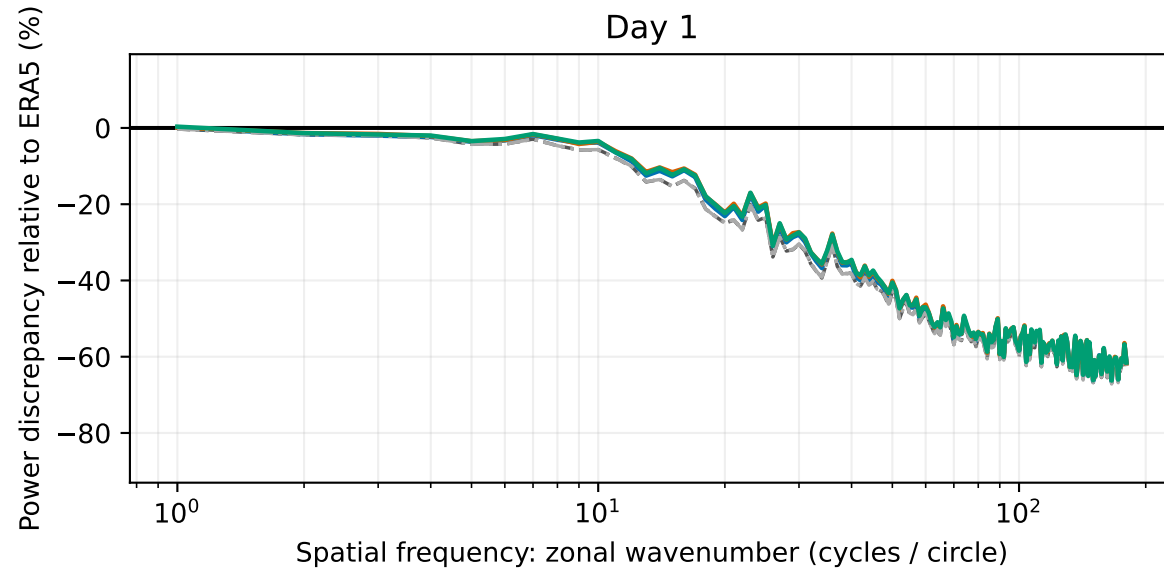


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
- - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
— di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

U10: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power

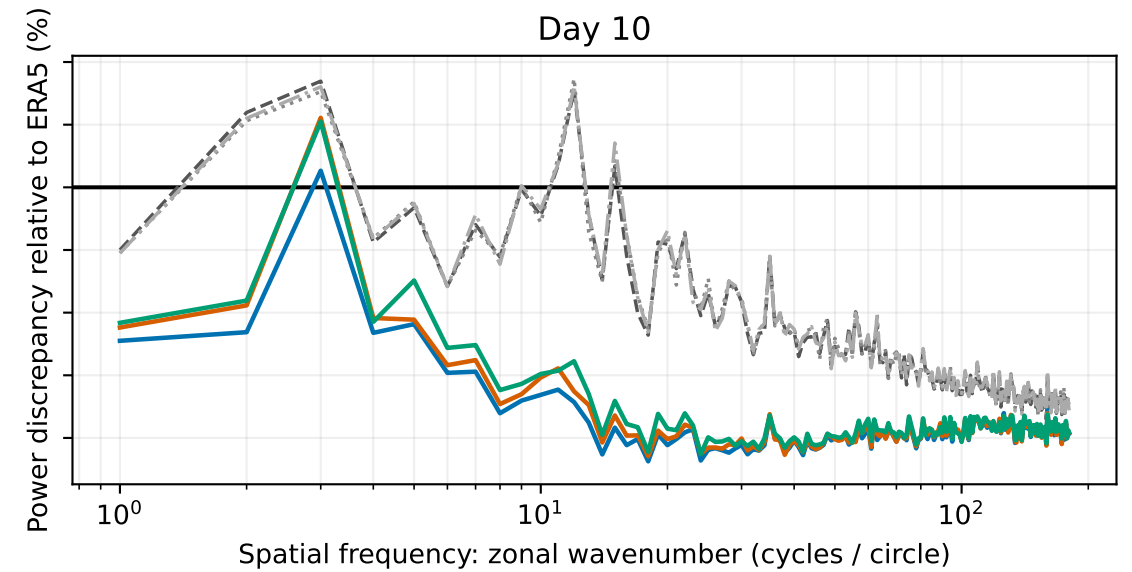
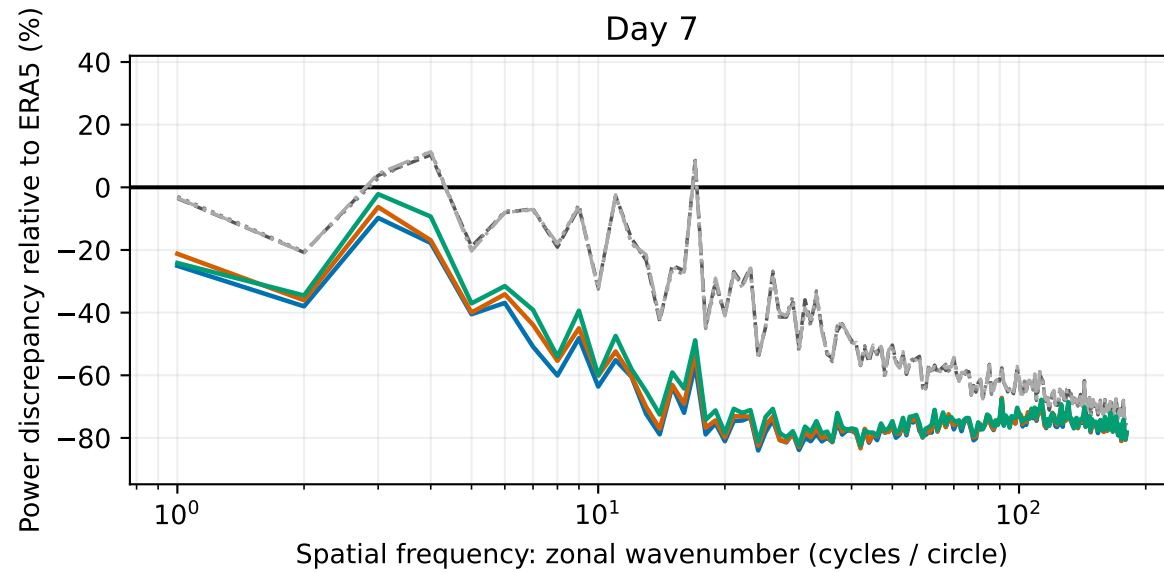
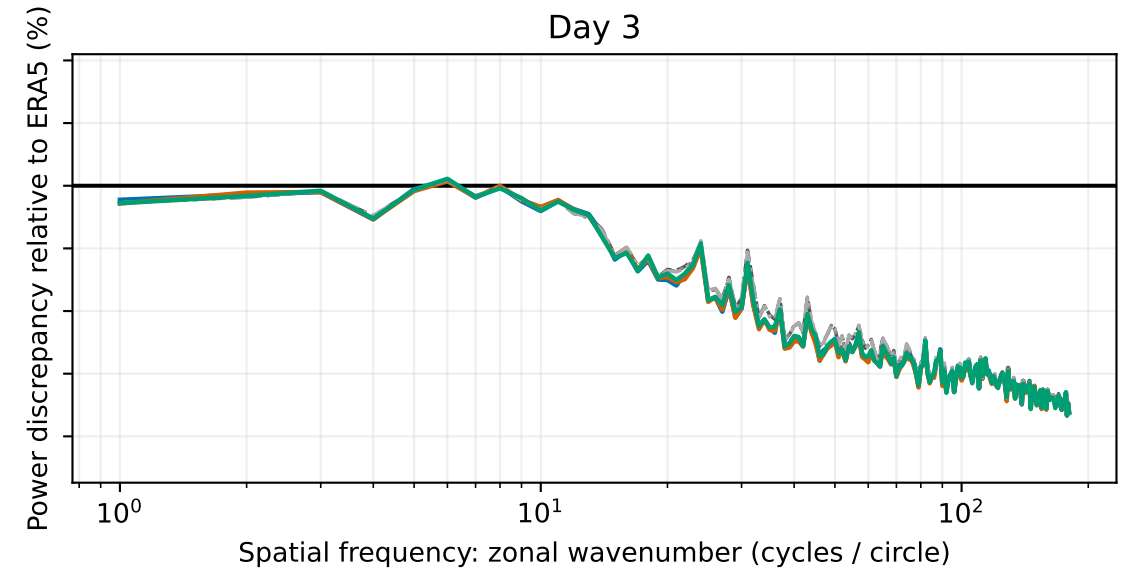
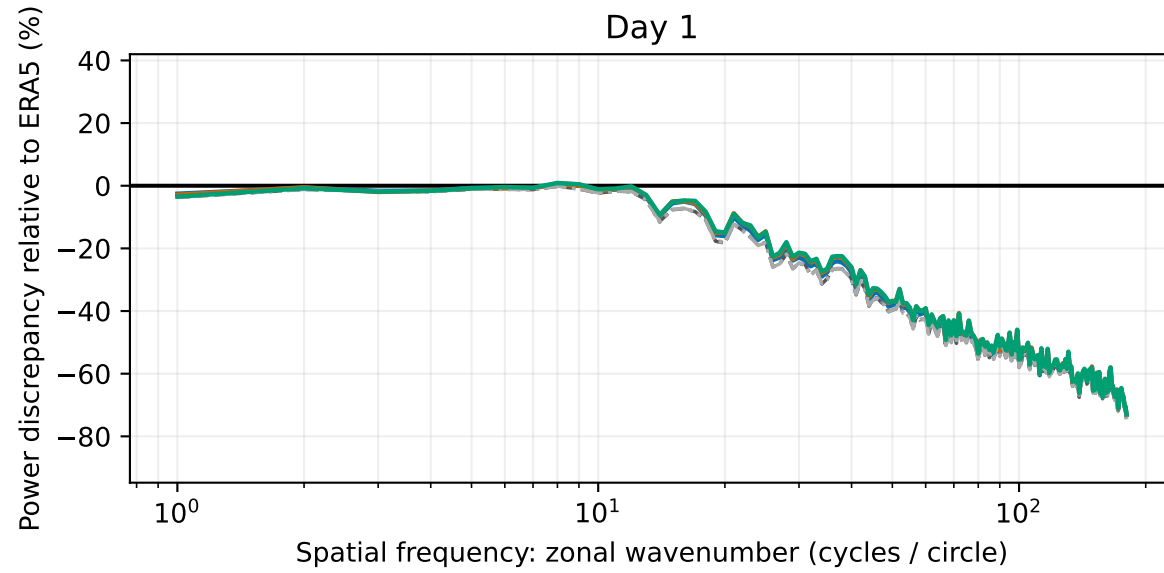


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
 Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

V10: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power



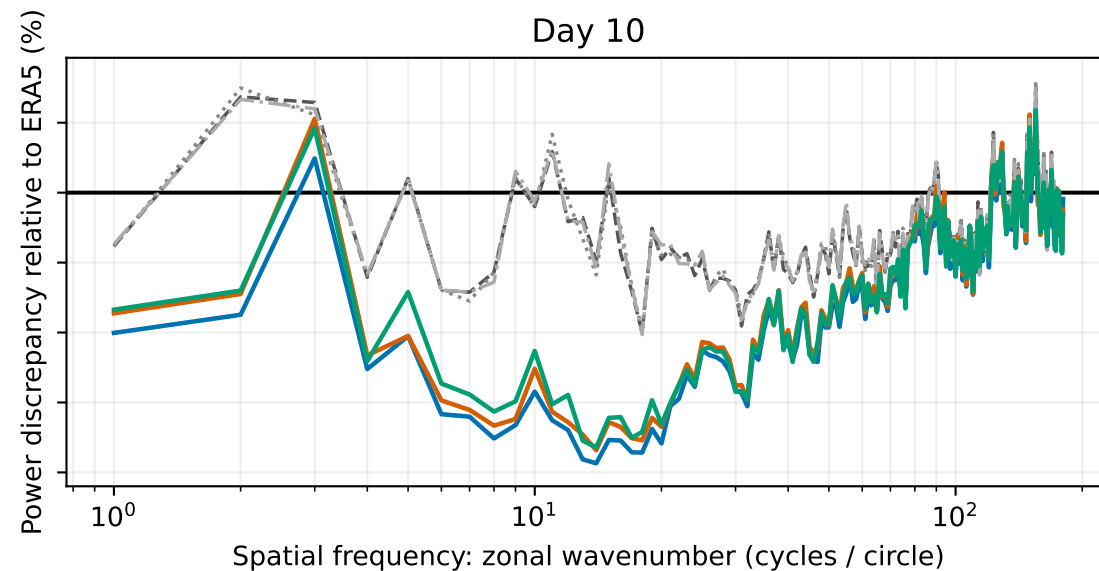
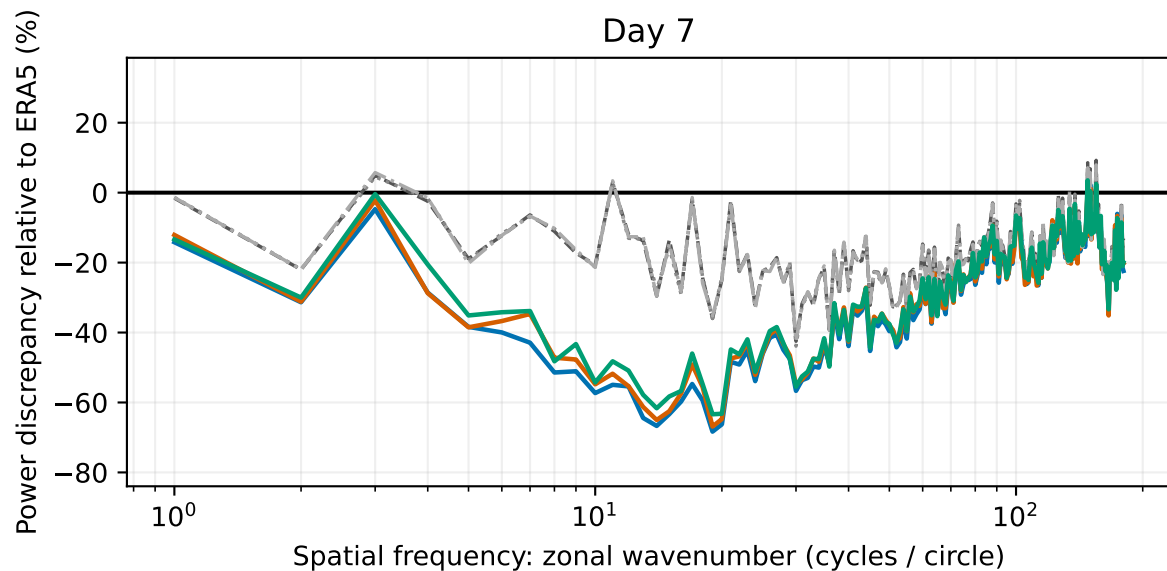
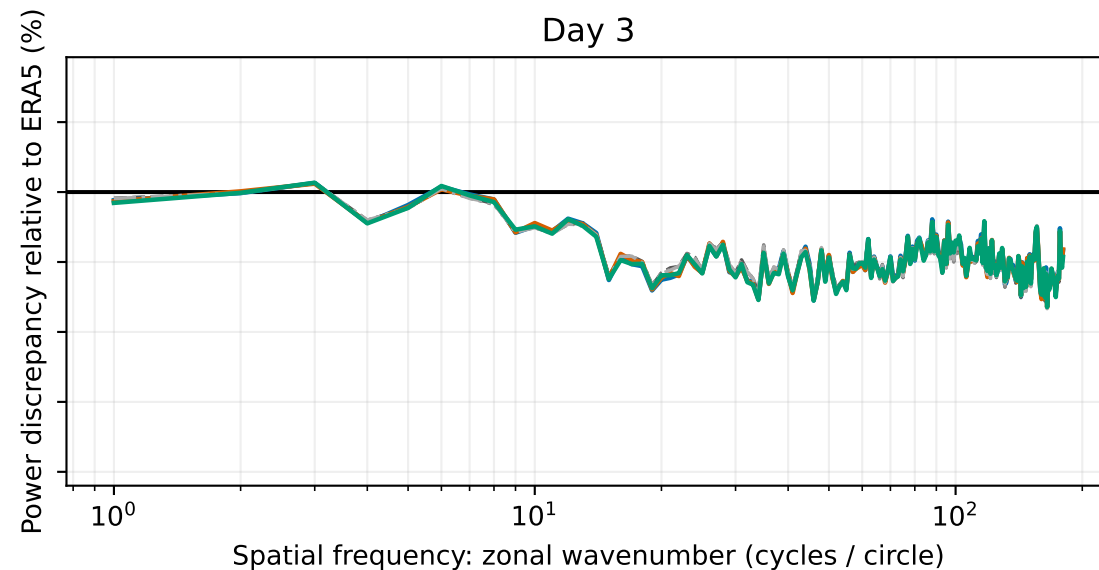
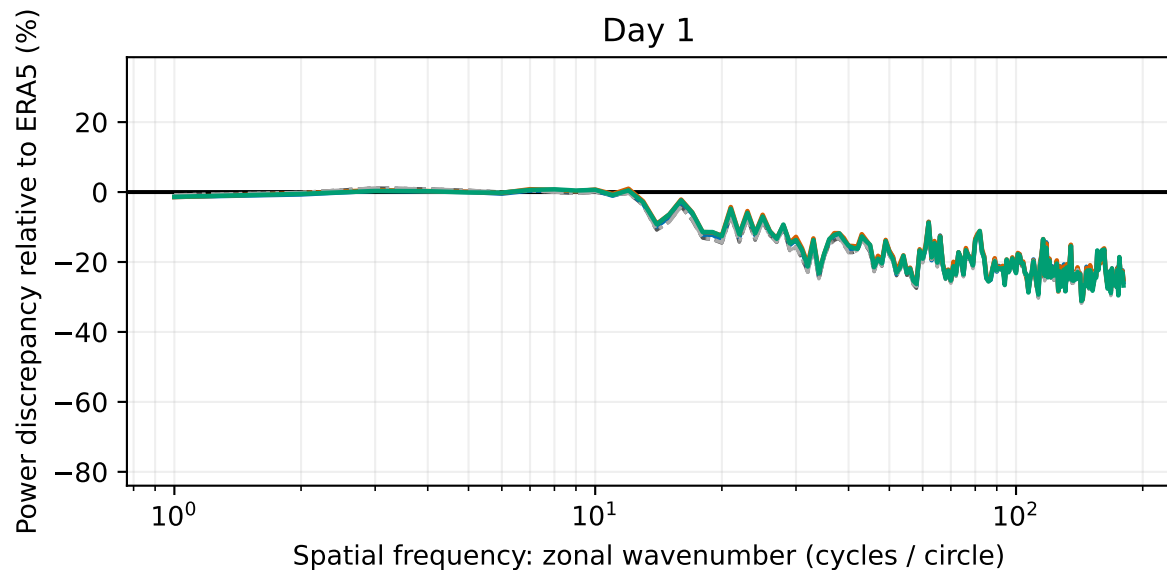
— ERA5: zero discrepancy ····· Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios

Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

MSLP: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power

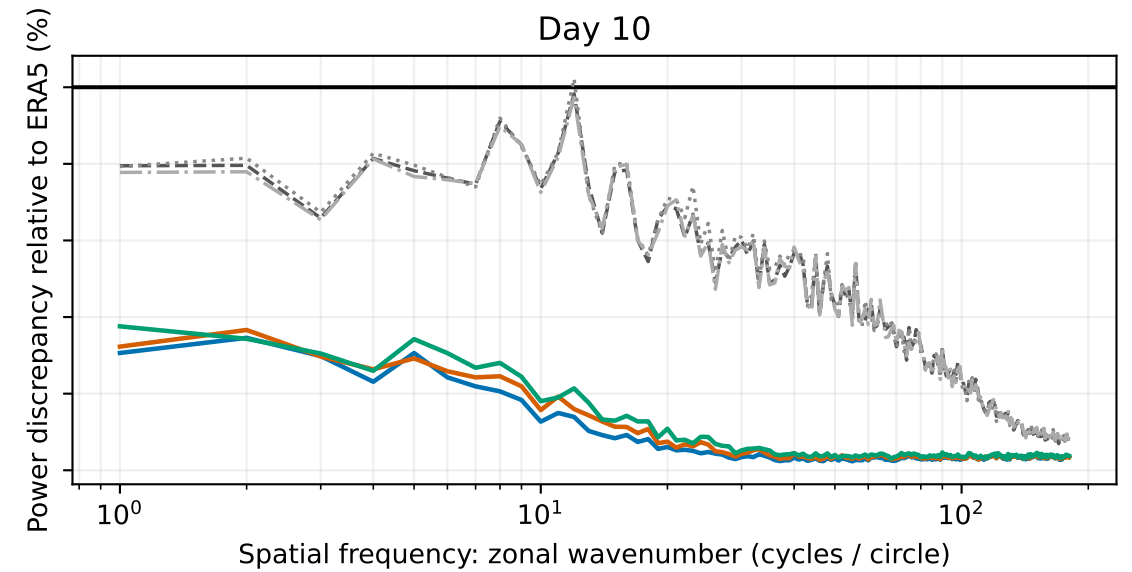
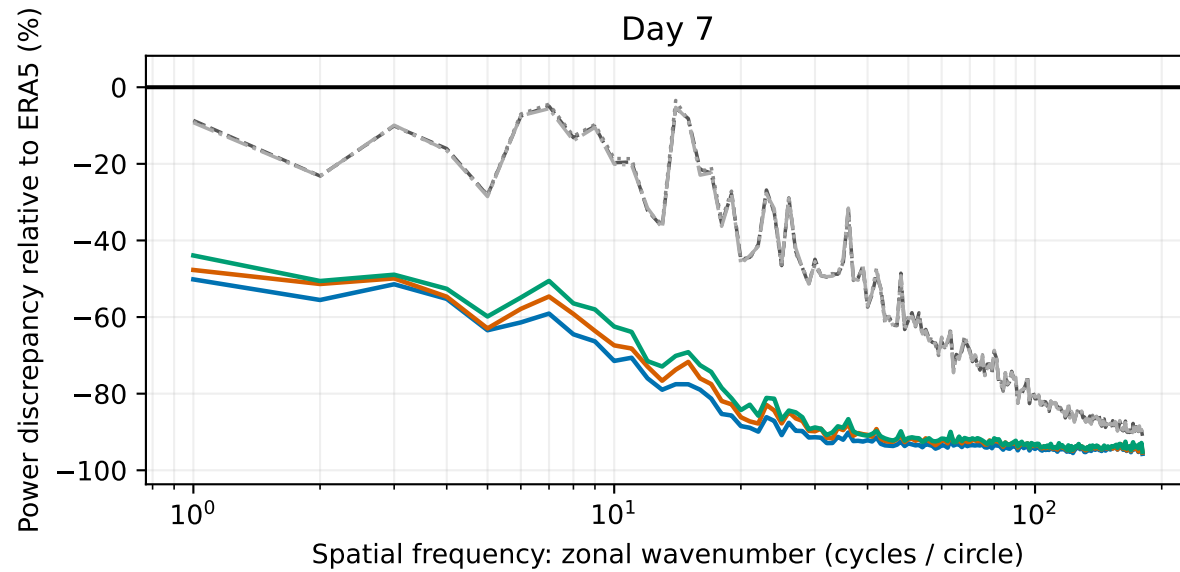
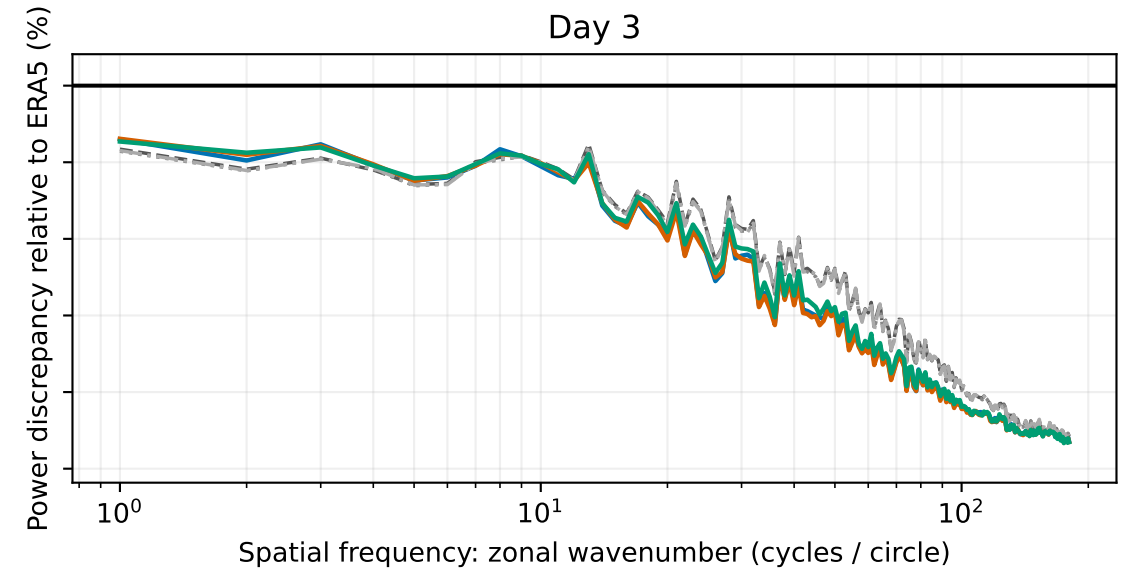
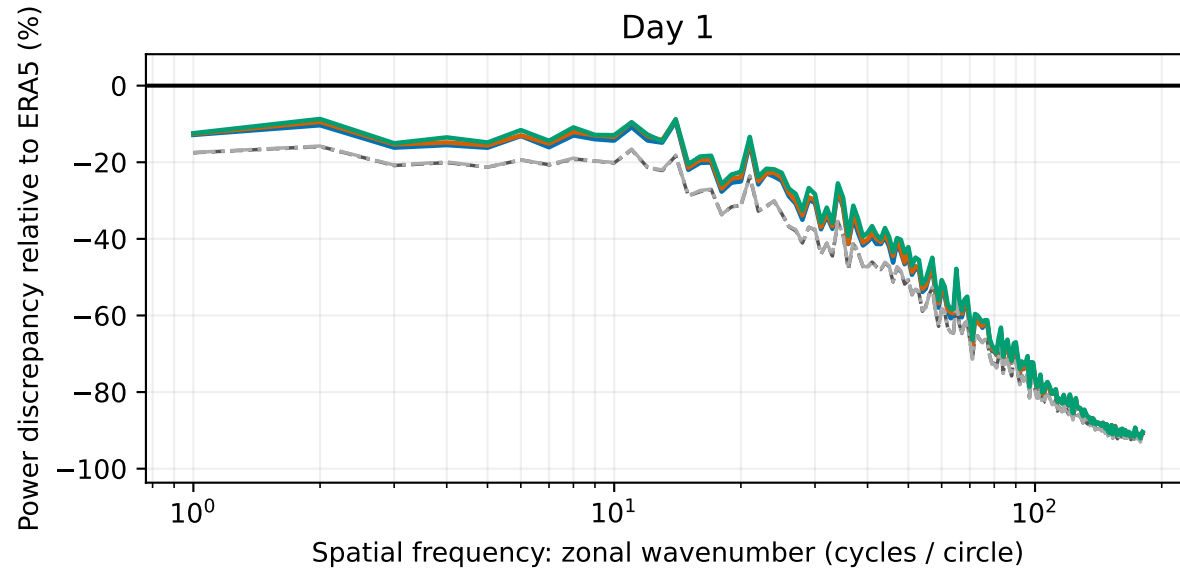


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
- - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
— di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

TP6: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power

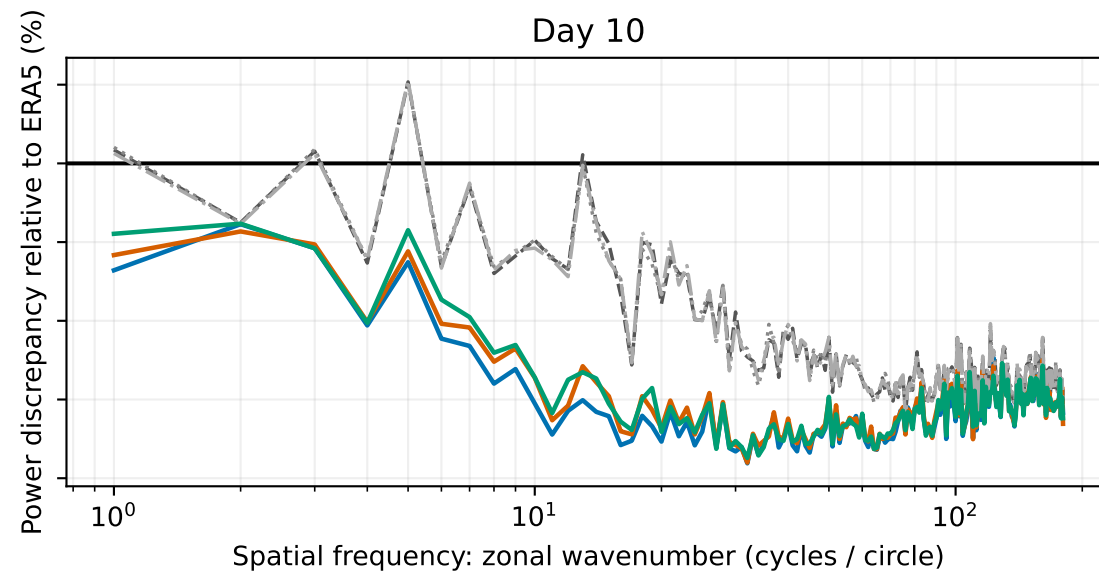
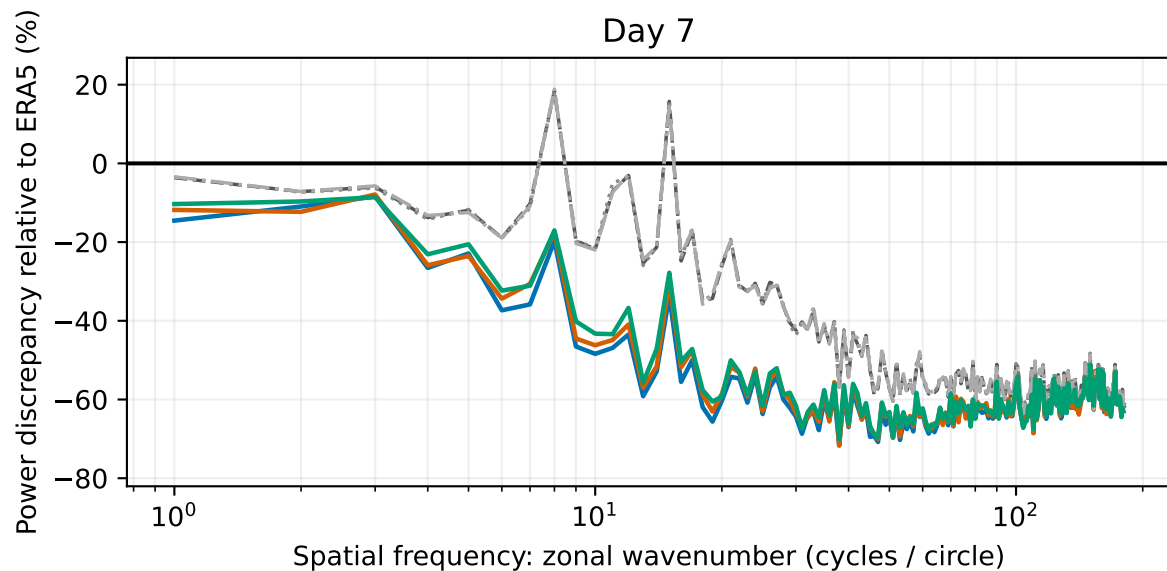
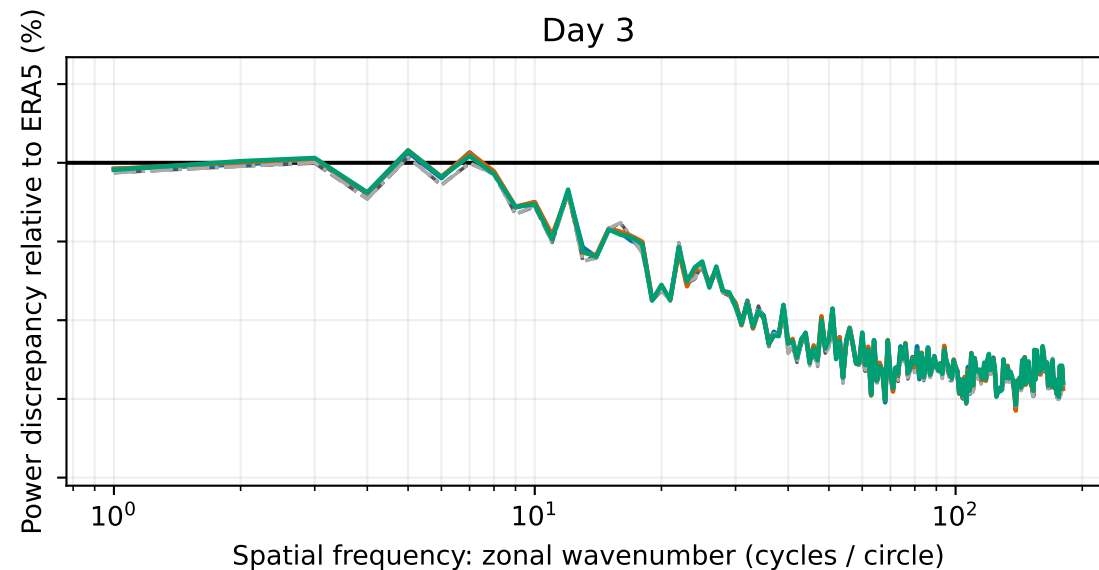
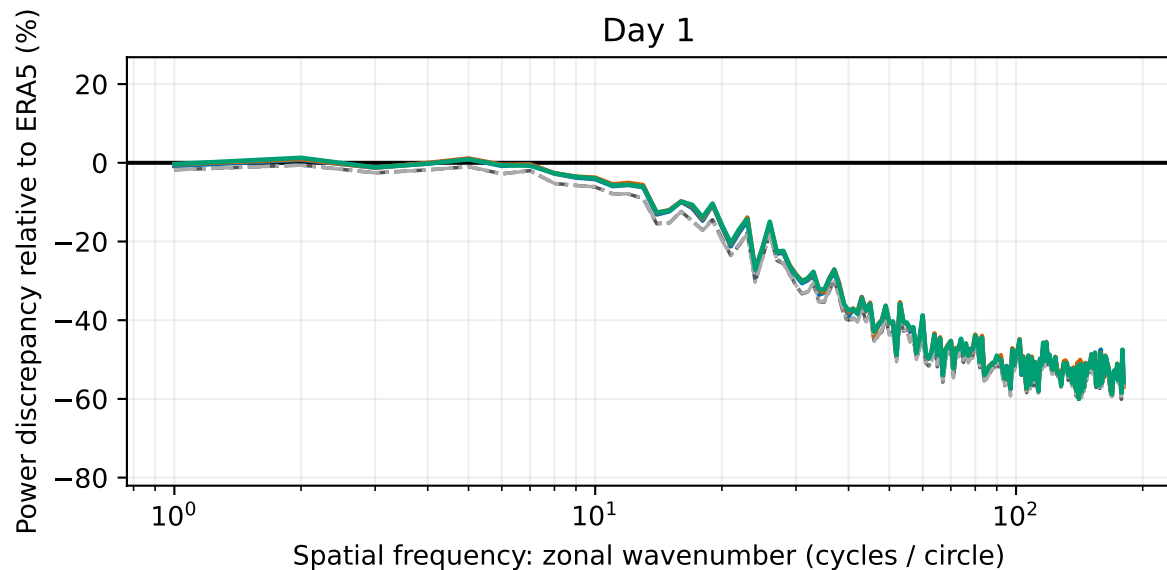


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
 Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

T850: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power

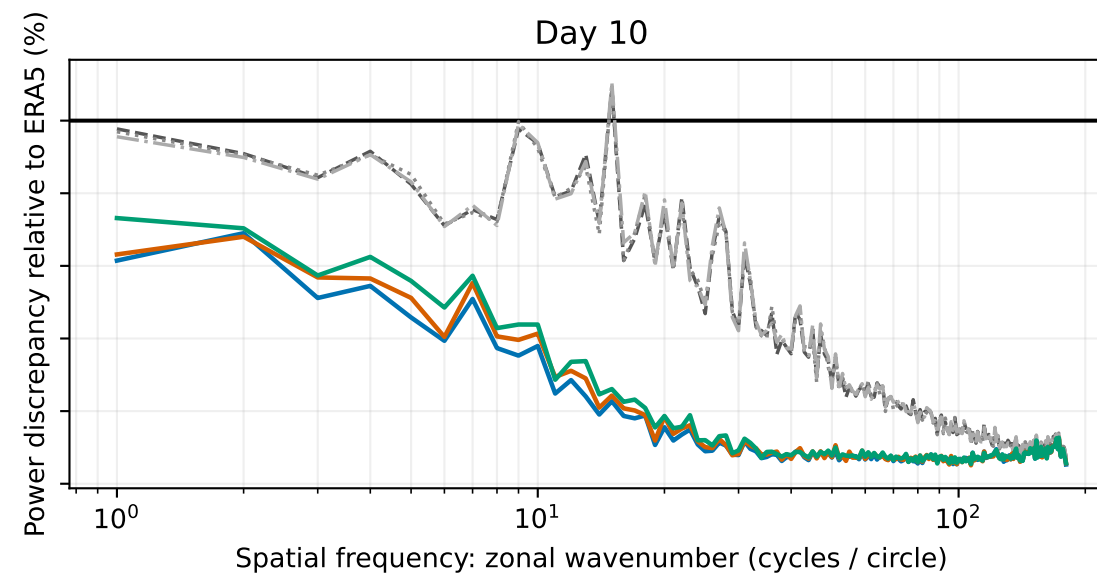
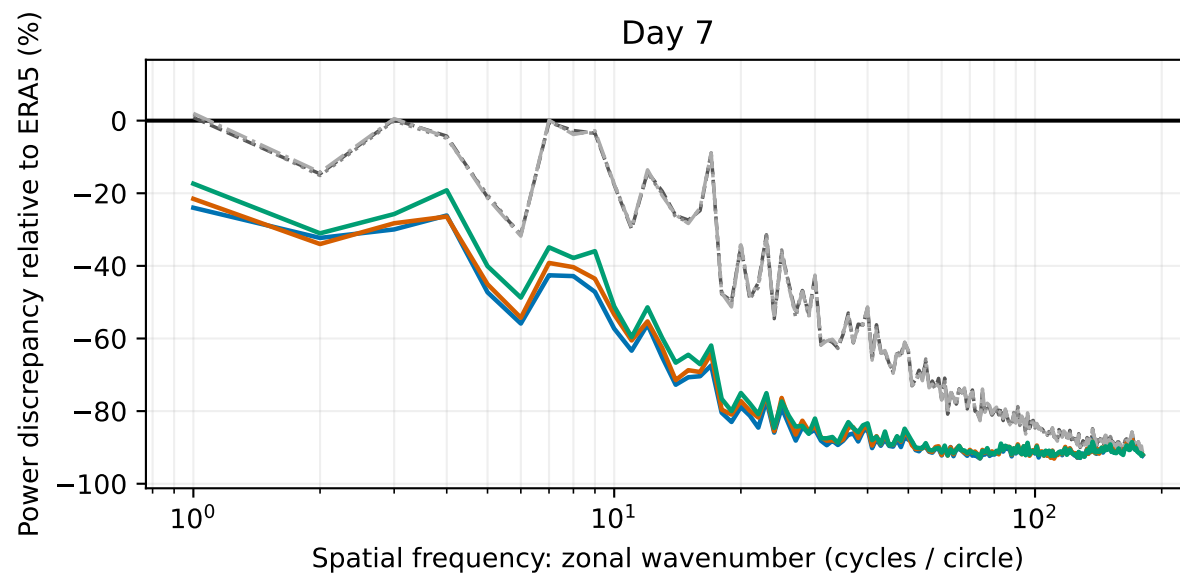
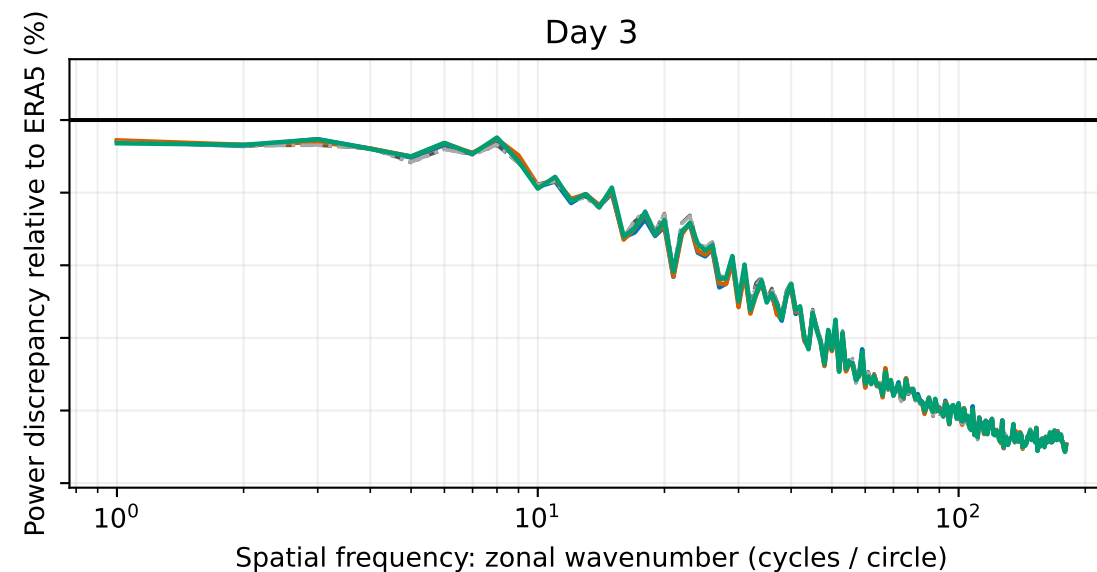
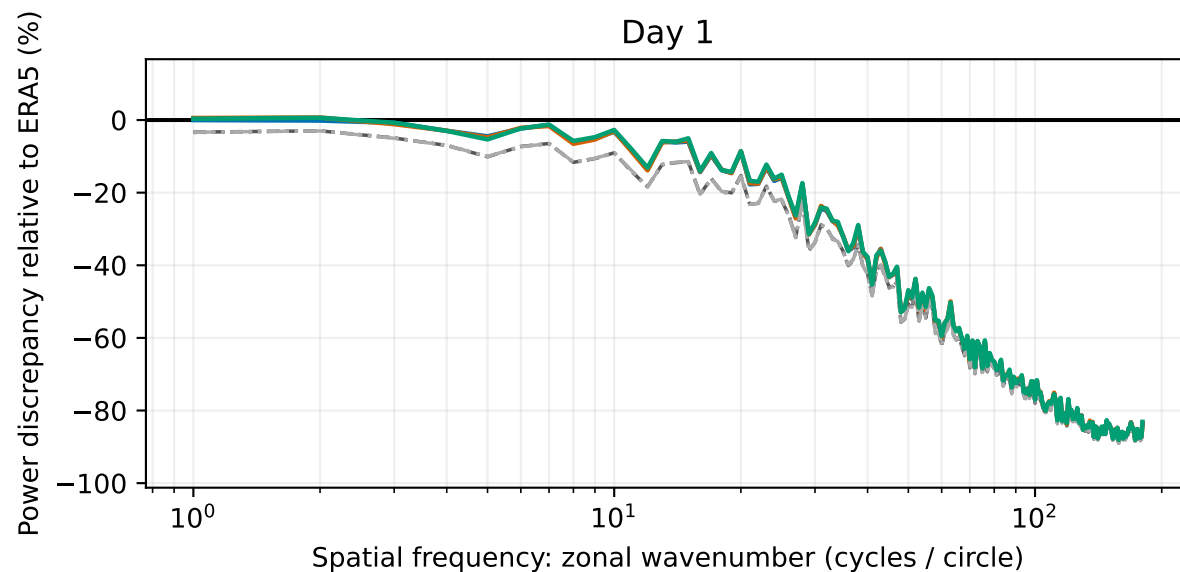


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
 Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

Q700: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power

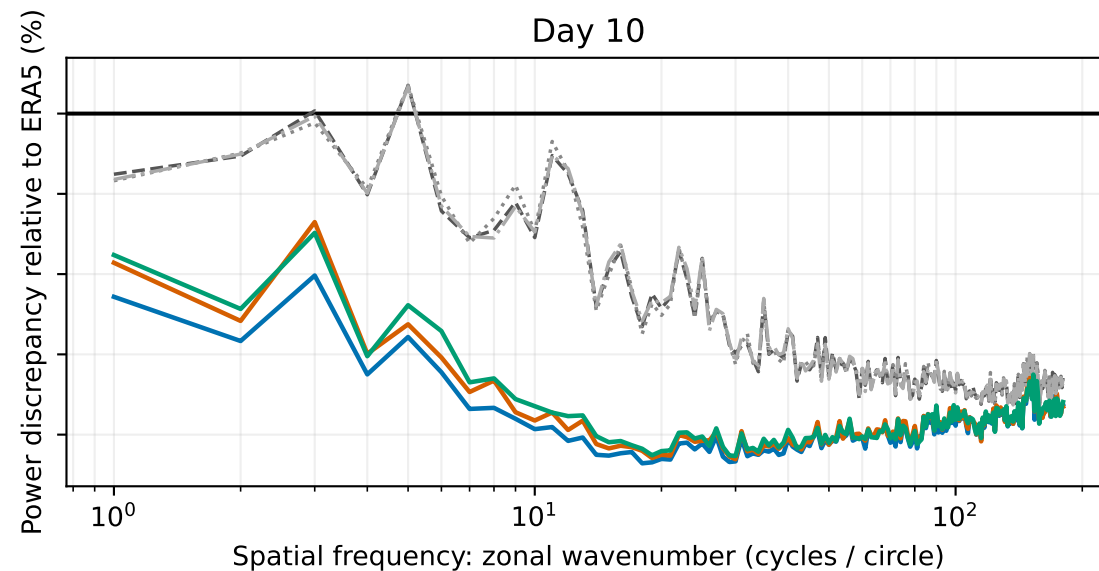
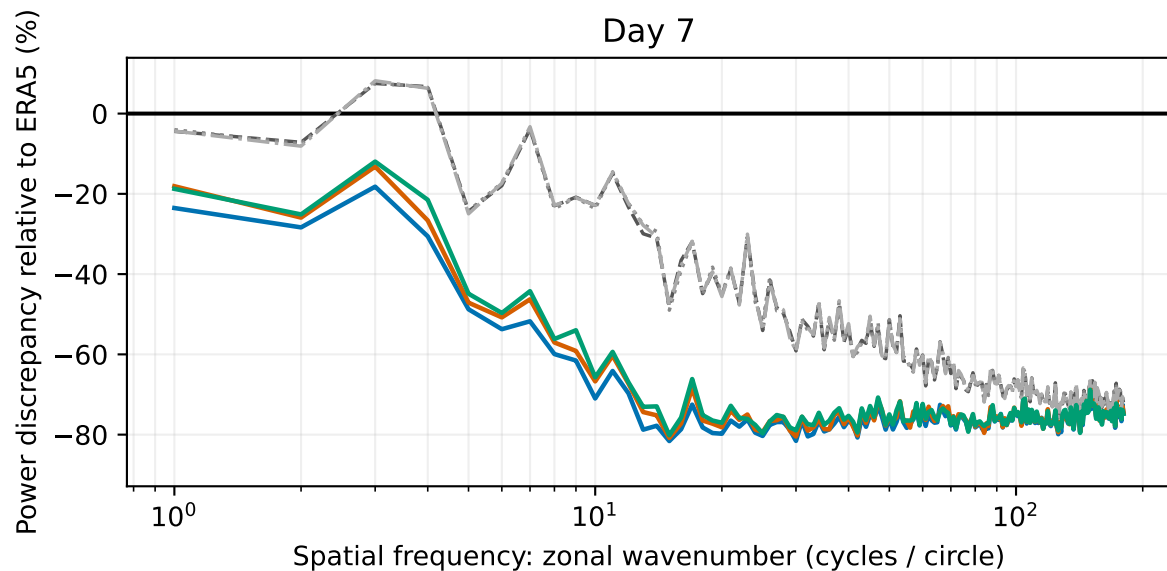
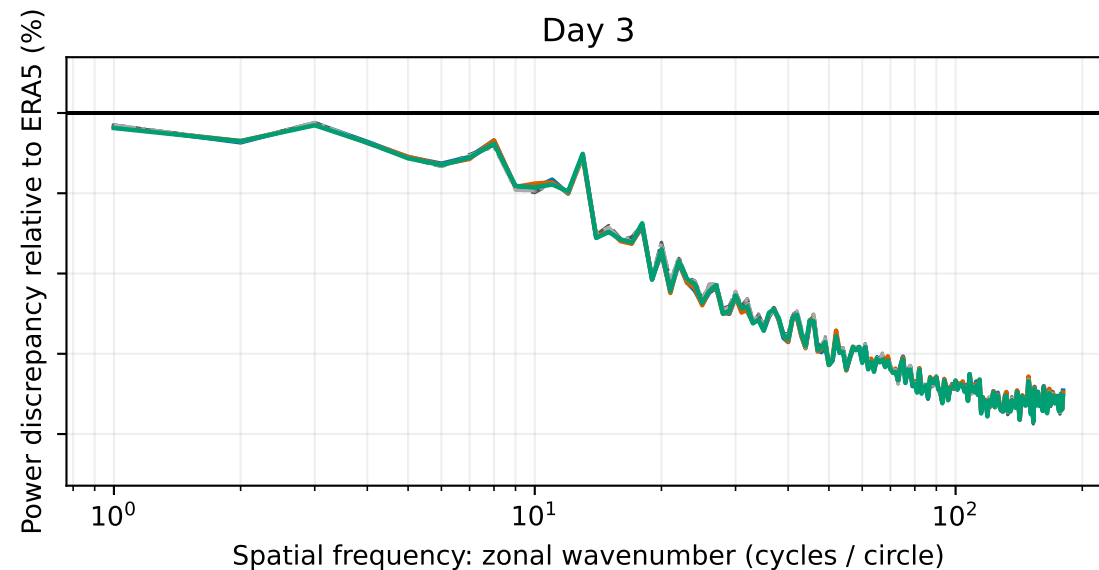
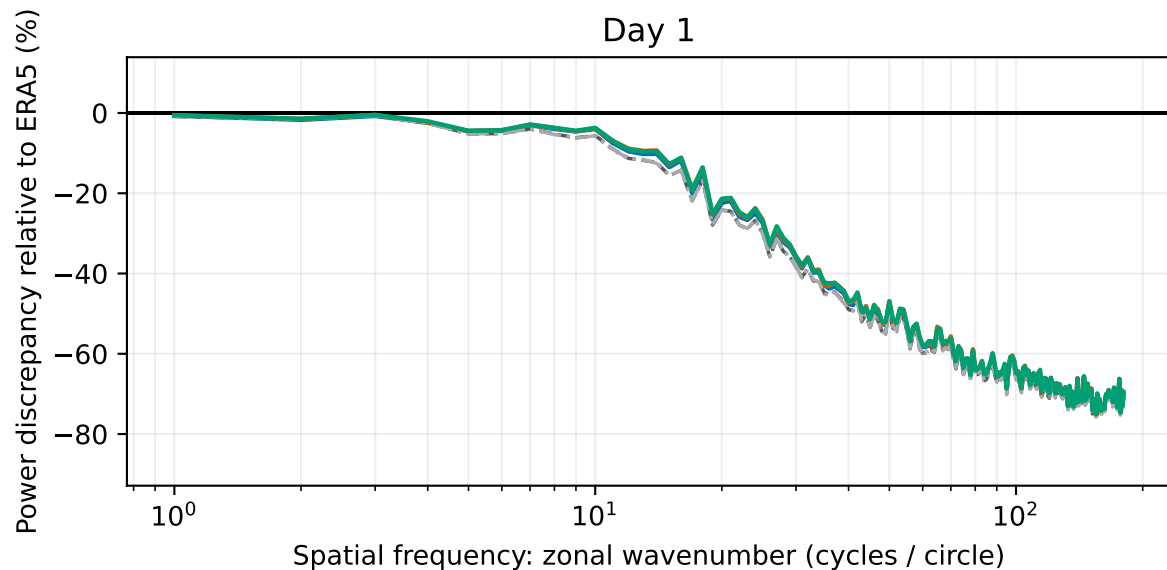


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
 Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

U850: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power

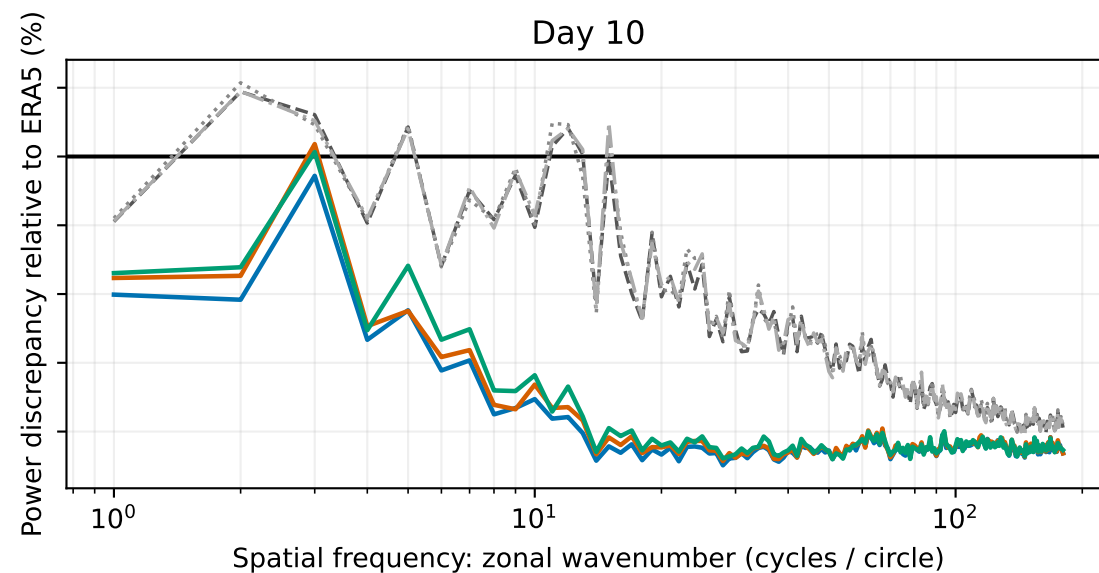
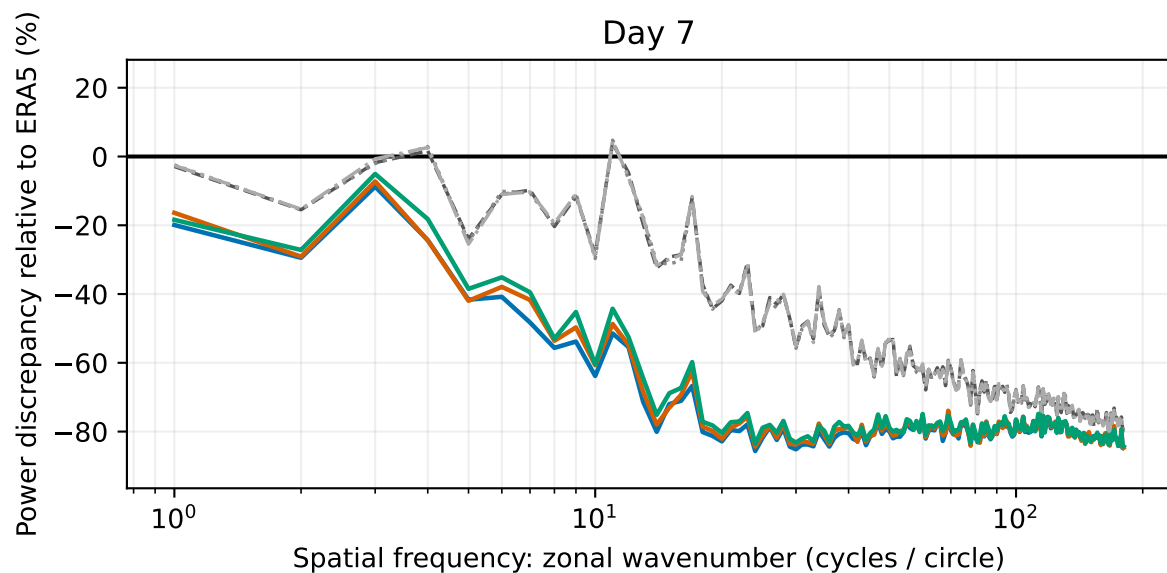
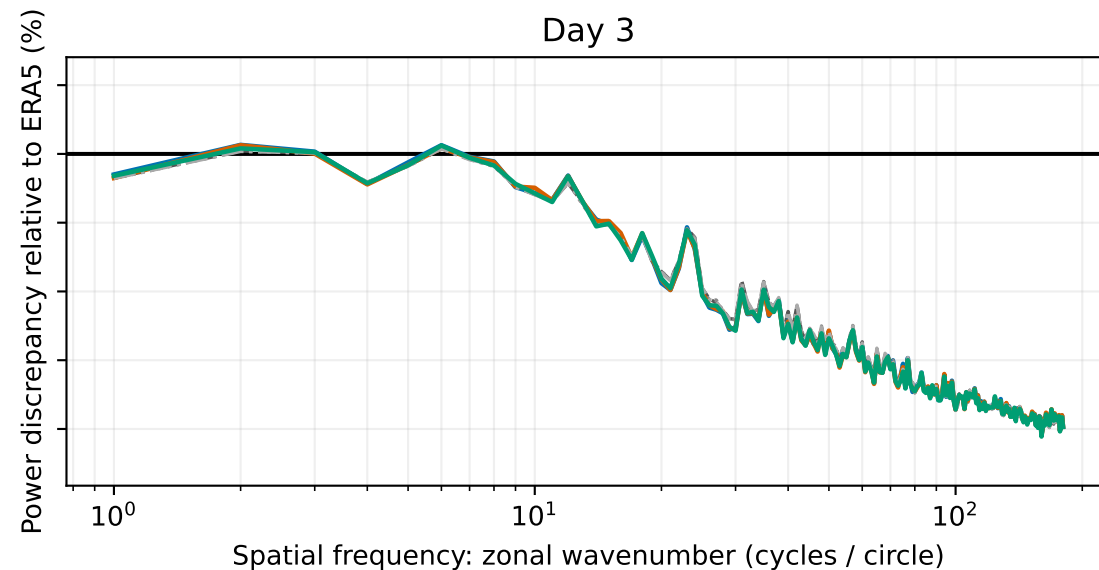
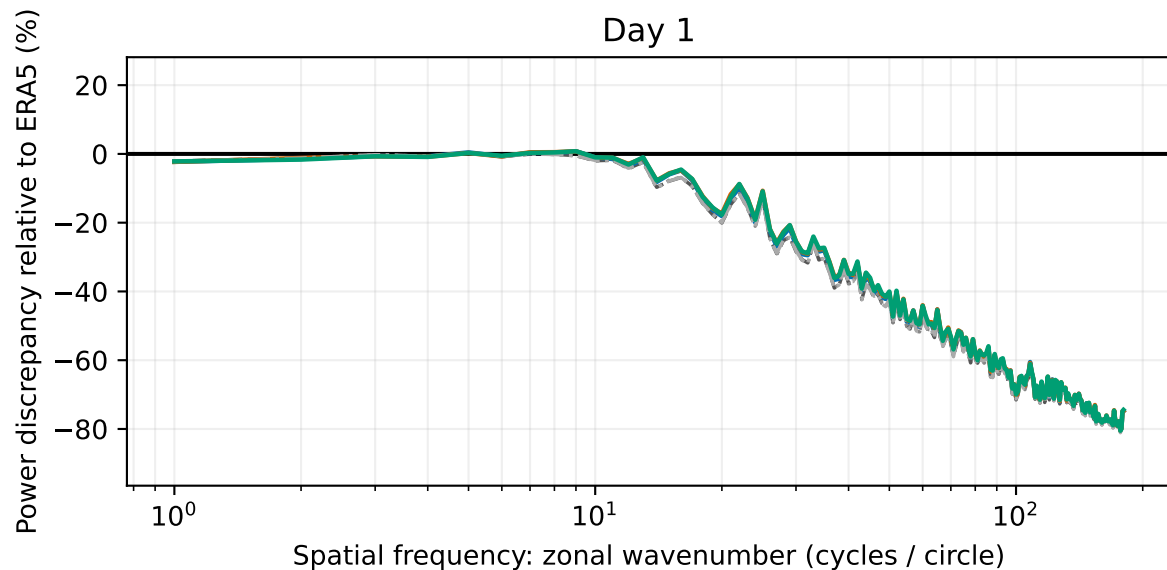


— ERA5: zero discrepancy ····· Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
 Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

V850: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power

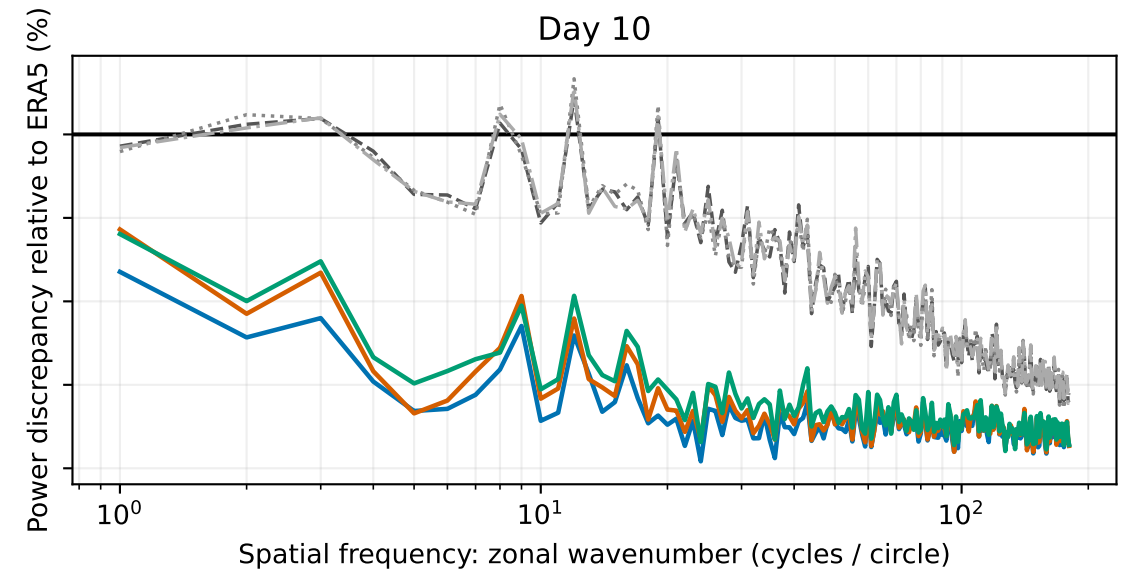
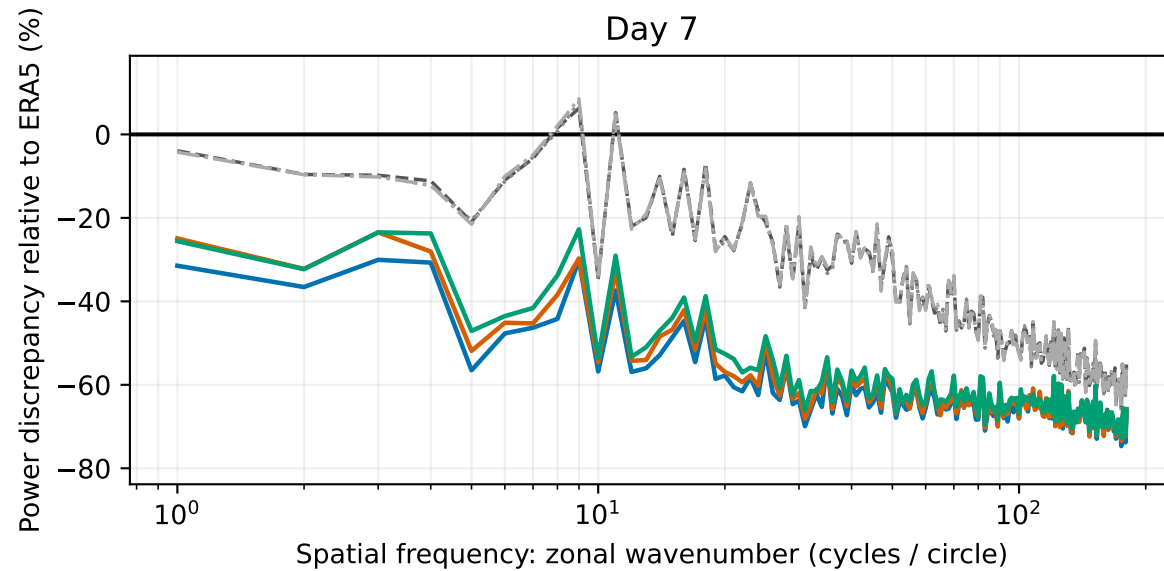
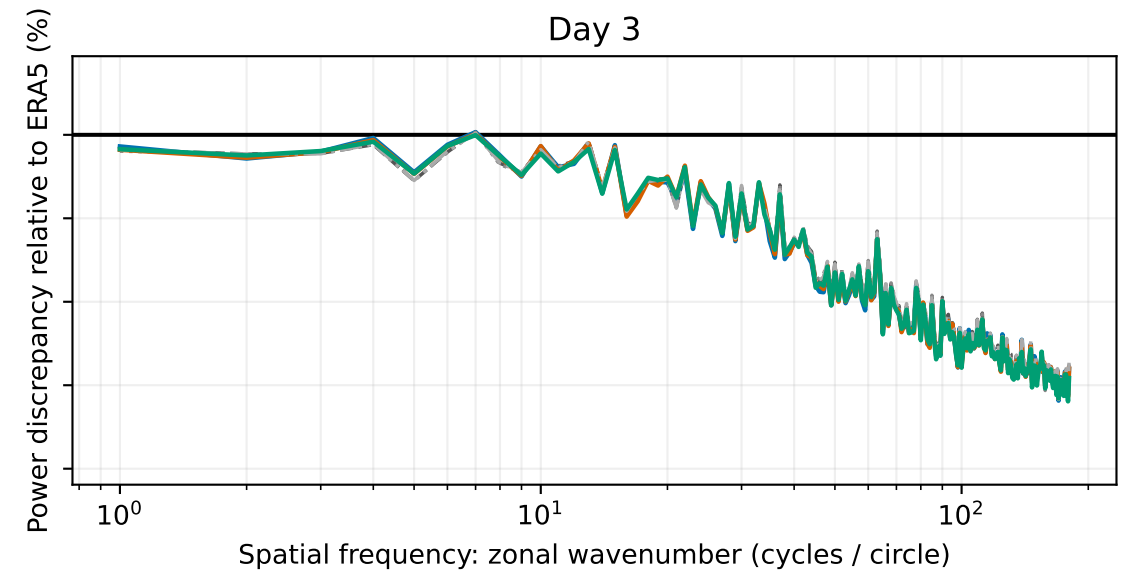
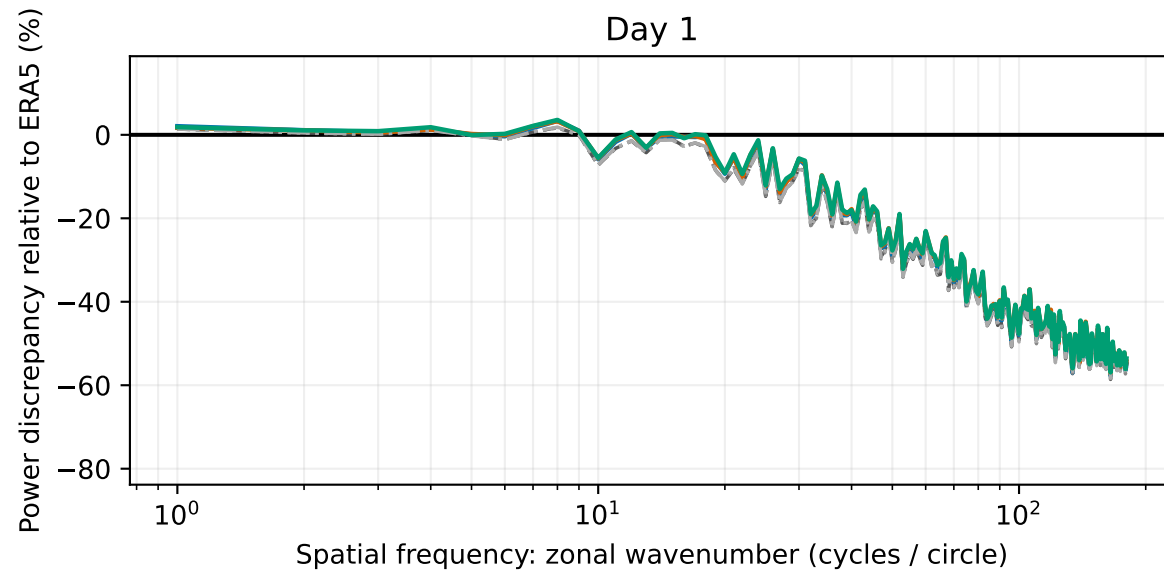


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
 Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

WS10: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power

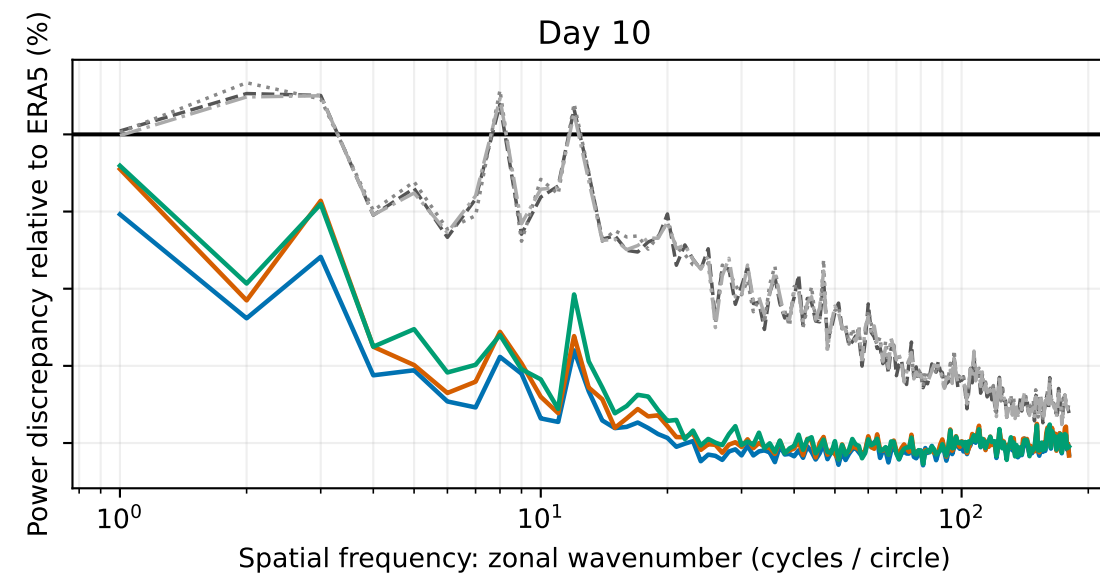
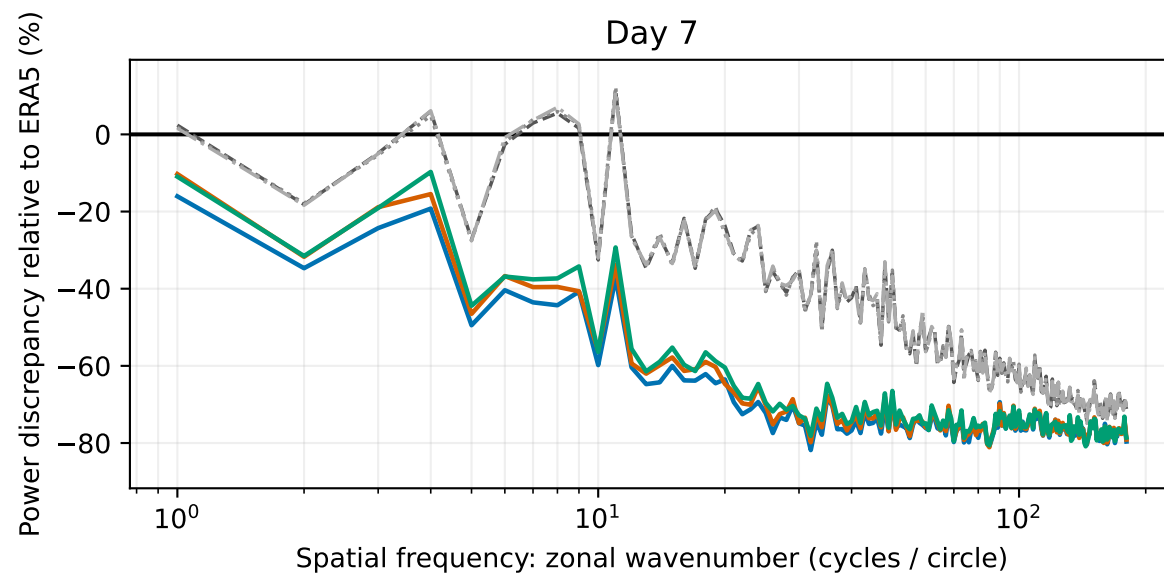
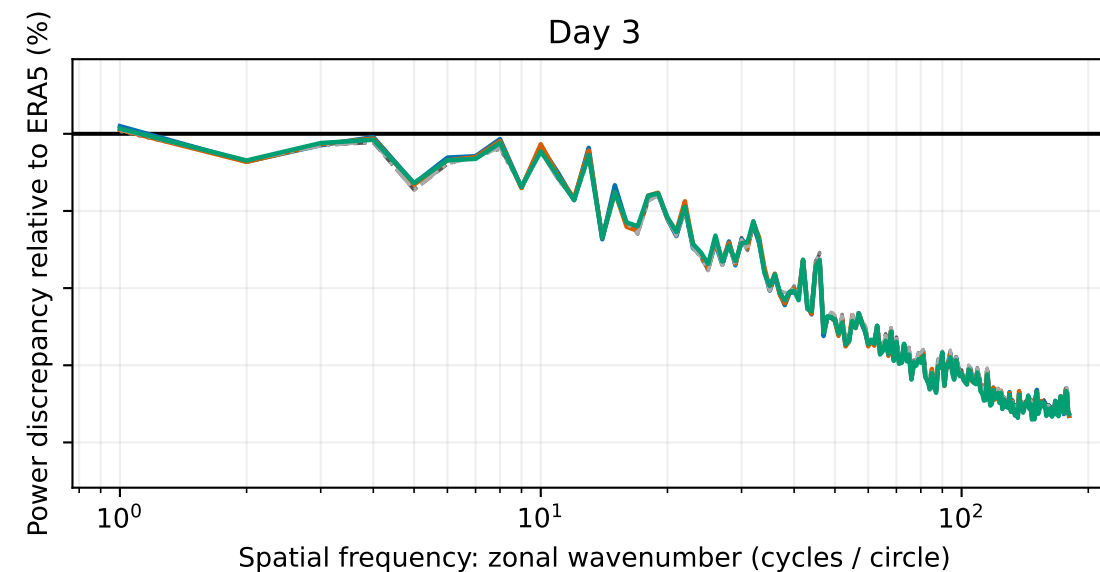
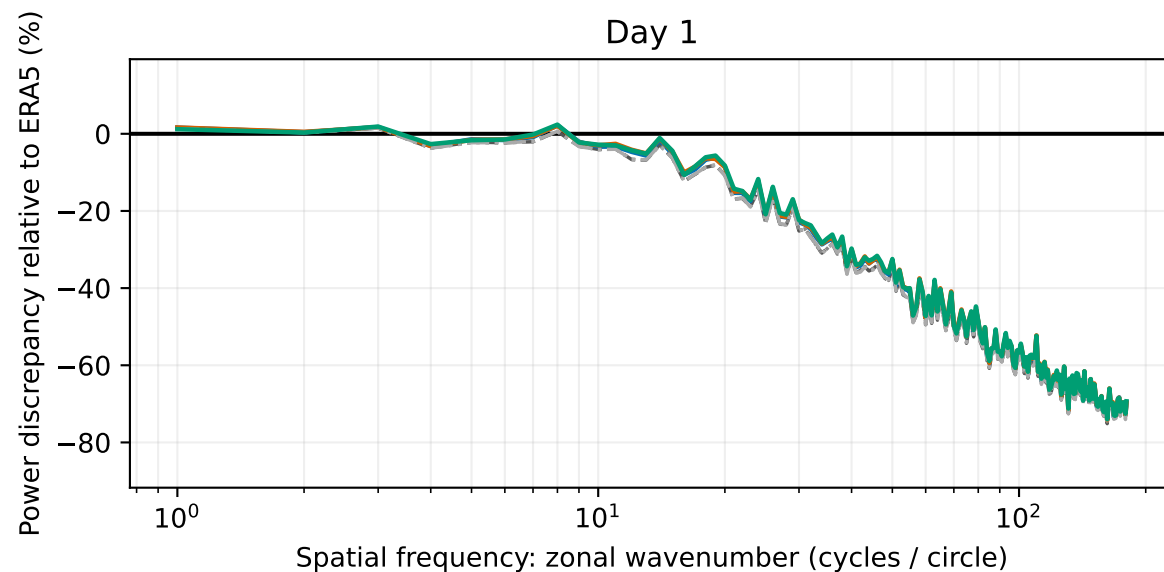


— ERA5: zero discrepancy ····· Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
 Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

WS850: power-spectrum discrepancy versus spatial frequency

$100 \times (\text{forecast power} / \text{ERA5 power} - 1)$ · Zero is ideal · Negative: missing power · Positive: excess power



— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
- - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
— di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.